

CSA1717-ARTIFICIAL INTELLIGENCE
PRACTICAL LAB EXPERIMENTS
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FORWARD CHAINING

Aim:

To implement the Forward Chaining algorithm in Prolog.

Objective:

To derive new facts from an initial set of facts by repeatedly applying rules.

Algorithm:

- 1. Start.**
- 2. Define initial facts.**
- 3. Define rules.**
- 4. Check whether all conditions of a rule are satisfied.**
- 5. If satisfied, derive the conclusion.**
- 6. Add the new fact to the knowledge base.**
- 7. Repeat until no new facts can be derived.**
- 8. Stop.**

Flowchart:

START

|

v

Load Initial Facts

|

v

Check Rules

|

v

Conditions Satisfied?

/ \

Yes No

| |

Derive Fact Check Next Rule

|

v

Add Fact to KB

|

v

New Fact Generated?

/ \

Yes No

| |

Repeat STOP

Prolog Program:

% Forward Chaining

fact(fever).

fact(cough).

rule([fever, cough], flu).

rule([flu], doctor_visit).

rule([doctor_visit], treatment).

forward :-

**collect_facts(Facts),
 forward_chain(Facts, FinalFacts),
 write('Final Facts: '),
 write(FinalFacts),
 nl.**

collect_facts(Facts) :-

findall(Fact, fact(Fact), Facts).

forward_chain(Facts, FinalFacts) :-

**findall(
 Conclusion,**

```

(
    rule(Conditions, Conclusion),
    all_present(Conditions, Facts),
    \+ member(Conclusion, Facts)
),
NewFacts
),
(
    NewFacts = [] ->
    FinalFacts = Facts
;
    append(Facts, NewFacts, UpdatedFacts),
    forward_chain(UpdatedFacts, FinalFacts)
).

```

all_present([], _).

all_present([H|T], Facts) :-
 member(H, Facts),
 all_present(T, Facts).

Query

?- forward.

Output

Final Facts: [fever,cough,flu,doctor_visit,treatment]

true.

Result:

Thus, Forward Chaining is successfully implemented to derive new facts from the knowledge base.